

Project title : State of the art climate prediction at a local scale

Supervisor : Emile Egreteau-Druet & Dr. Doreid Ammar

Groups: 2 (one will work on temperature and the other on precipitations)

Brief description :

One of today's major challenges is predicting the future climate in order to help organisations plan for it. The aim of this project is to use cutting-edge models to predict how the climate will evolve in France. In the context of global warming, we use global climate models (GCMs), which are physical models based on hypotheses about future sustainability policies. These results are then used by regional climate models (RCMs) to make predictions on a European scale. To obtain localised predictions for France, we employ bias correction and statistical downscaling methods. These methods correct errors in predictions from known periods and apply these corrections to future predictions. Bias corrections to historical data are mandatory for every climate model since they are the only way to ensure the validity of future predictions.

Starting from the results of the RCM, you will use the bias correction and statistical downscaling methods to compute predictions in an 8 km grid. This project will rely on a recent dataset that reflects the uncertainty in our knowledge about historical climate. First, you will use machine learning models such as k-nearest neighbours, random forest or support vector machines; then, you will use a model from a scientific publication.

Expected outcomes : Temperature and precipitation with the best resolution possible in France or one of its regions.

Required skills or prerequisites : Machine learning, Deep learning, need access to Aivancity's server or computation times will be unbearable

Project : A literature review and interactive dashboard on AI for healthcare compliance and regulations across countries

Supervisor: Dr. Anuradha Kar

Description:

Artificial Intelligence is increasingly used in healthcare for diagnostics, clinical decision support, and patient management. However, the adoption of AI is heavily influenced by compliance requirements, ethical frameworks, and regulatory landscapes that vary widely across countries. This project will conduct a systematic literature review of AI in healthcare regulations and compliance standards (e.g., FDA in the US, CE marking in the EU, HIPAA/GDPR for data privacy, local ministry guidelines).

The insights will be translated into an interactive dashboard that allows users to compare country-wise AI regulations in healthcare, track themes such as data governance, clinical validation, approval processes, and ethical considerations, and visualize global trends. The dashboard will serve as an accessible tool for researchers, policymakers, and industry professionals to understand and navigate regulatory diversity.

Objectives

1. **Literature Review** – Conduct a structured review of research papers, policy documents, and regulatory guidelines on AI in healthcare compliance.
2. **Thematic Synthesis** – Identify key themes such as data privacy, algorithmic transparency, validation standards, and approval mechanisms.
3. **Comparative Analysis** – Map regulatory similarities and differences across world regions (e.g., US, EU, Asia, Africa).
4. **Data Visualization** – Develop an interactive dashboard to present cross-country compliance requirements.
5. **Ethical Considerations** – Highlight gaps and challenges in balancing innovation with patient safety, privacy, and fairness.

Deliverables

1. **Literature Review Report** – A comprehensive document summarizing regulations, compliance frameworks, and thematic findings.
2. **Country-Wise Compliance Dataset** – Structured data compiled from literature and official guidelines.
3. **Interactive Dashboard Prototype** – A web-or PowerBI dashboard allowing users to:
 - Compare regulations across countries
 - Filter by themes (privacy, approval, ethics, safety)
 - Visualize global/regional trends in AI regulation

Required Skills / Prerequisites: Strong background in machine learning and deep learning. Introductory knowledge of compliance frameworks such as GDPR and the EU AI Act.

Project: Adaptive Learning Companion Agent: A Personalized AI Tutor for Dynamic Study Support

Supervisor: Dr. Anuradha Kar

Project Description

This project aims to design and implement an AI-powered adaptive learning companion agent that personalizes the learning journey for students. Unlike traditional Q&A chatbots, the agent will dynamically adjust learning content based on the learner's pace, weaknesses, and interests. It will generate personalized study plans, provide alerts for time management, and collaborate with other student agents to enable effective group project work.

Objectives

1. **Personalized Learning Pathways** – Develop models that adaptively curate content and exercises tailored to each learner's knowledge level, pace, and preferences.
2. **Real-Time Assessment** – Integrate NLP and ML techniques to evaluate student responses, identify weaknesses, and adjust difficulty levels dynamically.
3. **Behavioral Nudging** – Implement time management nudges, reminders, and motivational feedback using reinforcement learning and behavioral modeling.
4. **Collaborative Agent Framework** – Enable agents to communicate and share learning insights across students for **peer learning and group project support**.
5. **Human-AI Interaction** – Design an intuitive conversational interface that fosters engagement and trust in the agent.
6. **Ethics and Transparency** – Ensure explainability, fairness, and privacy in recommendations and content adaptation.

Deliverables

1. **Literature Review Report** – Survey of adaptive learning systems, intelligent tutoring agents, and agent collaboration in education.
2. **System Architecture Design** – Documentation and diagrams describing the multi-agent framework, NLP pipeline, and recommendation algorithms.
3. **Prototype AI Tutor** – A functional system capable of:
 - Adapting content to student performance and interests
 - Providing real-time personalized feedback
 - Sending time management suggestions
 - Supporting collaborative study tasks
4. **Evaluation Framework** – Metrics to assess learning gains, engagement, and usability (e.g., student satisfaction, retention rate, knowledge improvement).

Project title: Hybrid Cognitive Taxonomy Integration for Adaptive Quiz Generation in OneClickQuiz

Supervisor: Dr. Antoun Yaacoub

Project description: This project extends the integration of Bloom's Taxonomy and SOLO Taxonomy into OneClickQuiz (<https://oneclickquiz.fr>) by developing a hybrid classifier for generating adaptive multiple-choice questions (MCQs) in engineering domains, such as circuit design or programming. It addresses gaps in handling nuanced cognitive progressions under domain shifts, where single- taxonomy models (e.g., DistilBERT at 91% accuracy) may falter, by fusing taxonomies to enable real- time quiz adaptation that escalates cognitive demands based on learner responses. Inspired by recent advances in AI assessment frameworks, it hypothesizes a 12-18% alignment F1-score improvement.

Project objectives:

- Develop and fine-tune a multi-label DistilBERT ensemble for hybrid Bloom-SOLO classification, targeting 12-18% improvement in alignment F1-score over single-taxonomy baselines.
- Integrate the hybrid model into OneClickQuiz using prompt engineering for dynamic generation and adaptation.
- Evaluate causal impacts on engineering student learning outcomes through randomized controlled trials (RCTs) and mediation analysis on linguistic features like lexical density.

Expected outcomes:

- An enhanced OneClickQuiz plugin with hybrid taxonomy support and open-source codebase.
- A curated dataset of 2,000+ hybrid-labeled engineering MCQs.
- A technical report with empirical results, including mediation analysis, suitable for submission to IEEE Transactions on Learning Technologies or LAK 2026.

Required skills or prerequisites (if any): Machine learning (e.g., fine-tuning Transformers), Python programming, and familiarity with educational taxonomies.

Project title: Multimodal Quiz Generation: Incorporating Images and Code Snippets in OneClickQuiz for Engineering Domains

Supervisor: Dr. Antoun Yaacoub

Project description: This project enhances OneClickQuiz (<https://oneclickquiz.fr>) by adding multimodal generation capabilities, creating MCQs with images (e.g., circuit diagrams) or code snippets (e.g., Python algorithms) aligned to cognitive taxonomies. It fills gaps in engineering edtech where text-only quizzes limit spatial cognition, using vision-language models to hypothesize >90% taxonomy alignment F1 and 15% higher engagement versus text baselines.

Project objectives:

- Curate and label a multimodal dataset of 1,000+ engineering pairs, fine-tuning CLIP-DistilBERT for hybrid classification.
- Develop prompt-driven generation pipelines and integrate API hooks for media in OneClickQuiz.
- Conduct A/B user studies with engineering students to measure alignment accuracy, engagement (NPS), and cognitive load.

Expected outcomes:

- A multimodal prototype extension for OneClickQuiz, supporting image/code quizzes.
- A benchmark dataset of taxonomy-aligned multimodal engineering content.
- Comparative analysis report, targeting ICALT 2026 for publication.

Required skills or prerequisites (if any): Computer vision/NLP (e.g., CLIP integration), software engineering for plugins.

Project title: Ethical Prompt Optimization for Bias-Resilient Feedback in Multilingual Engineering Assessments

Supervisor: Dr. Antoun Yaacoub

Project description: Drawing from feedback linguistics and ethics in OneClickQuiz (<https://oneclickquiz.fr>), this project optimizes prompts for multilingual bias reduction (e.g., Eurocentric examples in global engineering), using constrained optimization to hypothesize 30% bias cuts without >5% readability trade-offs, addressing gaps in ethical multilingual AI for higher-order assessments.

Project objectives:

- Generate and annotate 1,800 multilingual feedbacks, fine-tuning mT5 with Lagrangian constraints for bias-readability.
- Develop a prompt optimizer and integrate into OneClickQuiz for cultural adaptation.
- Perform multilingual RCTs with bias surveys and performance regressions.

Expected outcomes:

- An ethical prompt optimizer toolkit for OneClickQuiz.
- A multilingual bias benchmark dataset.
- Trade-off analysis report, targeting FAccT 2026 or AIED 2026.

Required skills or prerequisites (if any): Multilingual NLP (mT5), optimization algorithms; ethical AI auditing.

Project title: Real-Time Adaptive Learning Paths in OneClickQuiz Using Reinforcement Learning

Supervisor: Dr. Antoun Yaacoub

Project description: Extending dynamic feedback mechanisms in OneClickQuiz (<https://oneclickquiz.fr>), this project implements reinforcement learning (RL) for personalized quiz paths that adapt in real-time to student responses, optimizing cognitive progression in engineering subjects (e.g., machine learning concepts). It addresses longitudinal gaps in RL for taxonomy alignment, hypothesizing 20% mastery gains via safe PPO agents with fairness rewards.

Project objectives:

- Build an RL environment (Gym) modeling taxonomy states and train a constrained PPO agent for path selection.
- Embed RL-driven adaptation into OneClickQuiz, with A/B testing for ethical constraints.
- Evaluate retention and progression through semester-long studies with mixed-effects models.

Expected outcomes:

- An RL module for adaptive paths in OneClickQuiz, with analytics dashboard.
- A longitudinal dataset of simulated and real student trajectories.
- Impact study report, suitable for EDM 2026 or Journal of Educational Data Mining.

Required skills or prerequisites (if any): Reinforcement learning (e.g., PPO algorithms), Python (Stable Baselines3); stats for longitudinal analysis.

Project title: Multi-Agentive AI with Self-Healing Capabilities for Adaptive Cognitive Alignment in OneClickQuiz

Supervisor: Dr. Antoun Yaacoub

Project description: We focus on enhancing OneClickQuiz (<https://oneclickquiz.fr>) with a multi-agentive AI system that integrates cognitive alignment (Bloom's and SOLO Taxonomies) with self-healing capabilities. Drawing from prior studies on feedback mechanisms and cognitive frameworks, the project introduces a network of specialized agents—e.g., a Question Generator Agent, Feedback Analyzer Agent, and Self-Healing Coordinator—collaborating to dynamically generate and refine engineering quizzes (e.g., thermodynamics or coding). These agents leverage a shared knowledge base and employ self-healing mechanisms to detect and correct alignment errors or system failures in real-time, addressing gaps in robust, adaptive educational AI as noted in 2025 reviews on resilient AI systems. The hypothesis is that multi-agent coordination with self-healing will improve alignment accuracy by 25% and system uptime by 15% over single-agent baselines.

Project objectives:

- Design and implement a multi-agent system using frameworks like A2C or OpenAI Gym Multi-Agent, with agents specialized for question generation, feedback analysis (building on RoBERTa-MTL), and self-healing (detecting misalignments via anomaly detection).
- Develop self-healing protocols using reinforcement learning (e.g., Q-learning for error recovery) and integrate them into OneClickQuiz to autonomously adjust agent behaviors or regenerate content when cognitive misalignment or system errors occur.
- Conduct a mixed-methods evaluation with 60 engineering students over a semester, assessing alignment F1-score (>93%), self-healing success rate (>85%), and user satisfaction (NPS > 7) through logs, surveys, and A/B testing against a single-agent version.

Expected outcomes:

- A multi-agentive, self-healing OneClickQuiz extension with open-source code, demonstrating collaborative AI for educational adaptability.
- A dataset of 2,500+ agent-generated and self-healed quiz-feedback pairs, annotated for cognitive alignment and error types.
- A research paper exploring multi-agent dynamics and self-healing in edtech, targeting IEEE Transactions on Learning Technologies or AIED 2026.

Required skills or prerequisites (if any): Multi-agent reinforcement learning, anomaly detection, Python (e.g., TensorFlow, PyTorch), and familiarity with cognitive taxonomies.

Project Title: Exploring Brain Signals and Affective Computing for Targeted Marketing and Consumer Behavioral Persuasion

Supervisor: Dr. Abdelhamid Mellouk

Project Description

At the intersection of cognitive neuroscience, artificial intelligence, and marketing, a new field of innovation is emerging—centered on analyzing mental and emotional states through biological and brain signals. The aim is to better understand and even predict consumer reactions to advertising stimuli or shopping environments, thereby optimizing personalization and the effectiveness of marketing messages. This domain, often referred to as *augmented neuromarketing* or *adaptive cognitive marketing*, relies on data from electroencephalography (EEG), facial expressions, skin conductance, heart rate signals, and eye-tracking. These signals are processed using affective computing models, capable of estimating in real time an individual's attention, cognitive load, interest, or emotional state.

Project Objectives

- Identify cognitive and emotional correlates of consumer responses to marketing content using large-scale data processing tools.
- Develop and/or test automated models for analyzing brain and physiological signals in consumer contexts.
- Investigate the effects of emotionally personalized advertising messages on consumer decision-making.
- Explore the ethical implications of such techniques, including consent, potential manipulation, and the protection of sensitive data.

Expected Outcomes

- A validated framework for linking brain/physiological signals with consumer emotional and cognitive responses.
- Prototypes or proof-of-concept models for real-time analysis of EEG and other biosignals in marketing scenarios.
- Insights into the effectiveness of emotionally adaptive marketing strategies.
- A critical evaluation of ethical considerations and recommendations for responsible use of affective computing in marketing.

Required Skills

- Knowledge of artificial intelligence and machine learning (including LLMs and signal processing algorithms). Understanding of consumer behavior and targeted marketing strategies. Awareness of data privacy and ethical issues related to biometric and emotional data.

Project title: Analysis and Processing of Market Dynamics and Arbitrage Opportunities in High-Frequency Trading Using High-Resolution Financial Data

Supervisor: Dr. Abdelhamid Mellouk

Project Description

High-frequency trading (HFT) uses sophisticated algorithms to exploit high-frequency financial data, including order books, real-time quotation streams, and market signals. The quality, granularity, and processing speed of financial data are critical for detecting arbitrage opportunities, optimizing market-making or scalping strategies, and understanding market microstructure. However, several challenges remain, including technical (real-time storage and processing, latency), algorithmic, economic and ethical (fairness among market actors).

This project focuses on the pre-processing stage of financial data—a critical step often overlooked. The project aims to develop methods for real-time data pre-processing using generative AI, machine learning, and statistical tools to detect inconsistencies and noise in massive financial data streams.

Project Objectives

- Design and implement algorithms for efficient pre-processing and filtering of high-frequency financial data.
- Detect inconsistencies and anomalies in real-time trading data streams.
- Improve decision latency to optimize algorithmic trading performance.
- Explore the role of AI (including generative AI and LLMs) in enhancing data quality for trading strategies. Assess the technical, economic, and ethical challenges associated with high-frequency trading data.

Expected Outcomes

- A prototype framework for real-time cleaning and pre-processing of high-frequency financial data. Improved execution speed and accuracy of trading algorithms through reduced decision latency.
- Benchmarking of machine learning and AI-based methods for data grooming in HFT contexts. Insights into ethical and fairness concerns in the use of premium and processed financial data.

Required Skills

- Strong knowledge of data science, machine learning, and statistical modeling. Familiarity with financial markets, trading strategies, and market microstructure. Experience with big data tools and high-frequency data processing techniques. Understanding of real-time systems, latency issues, and optimization methods.

